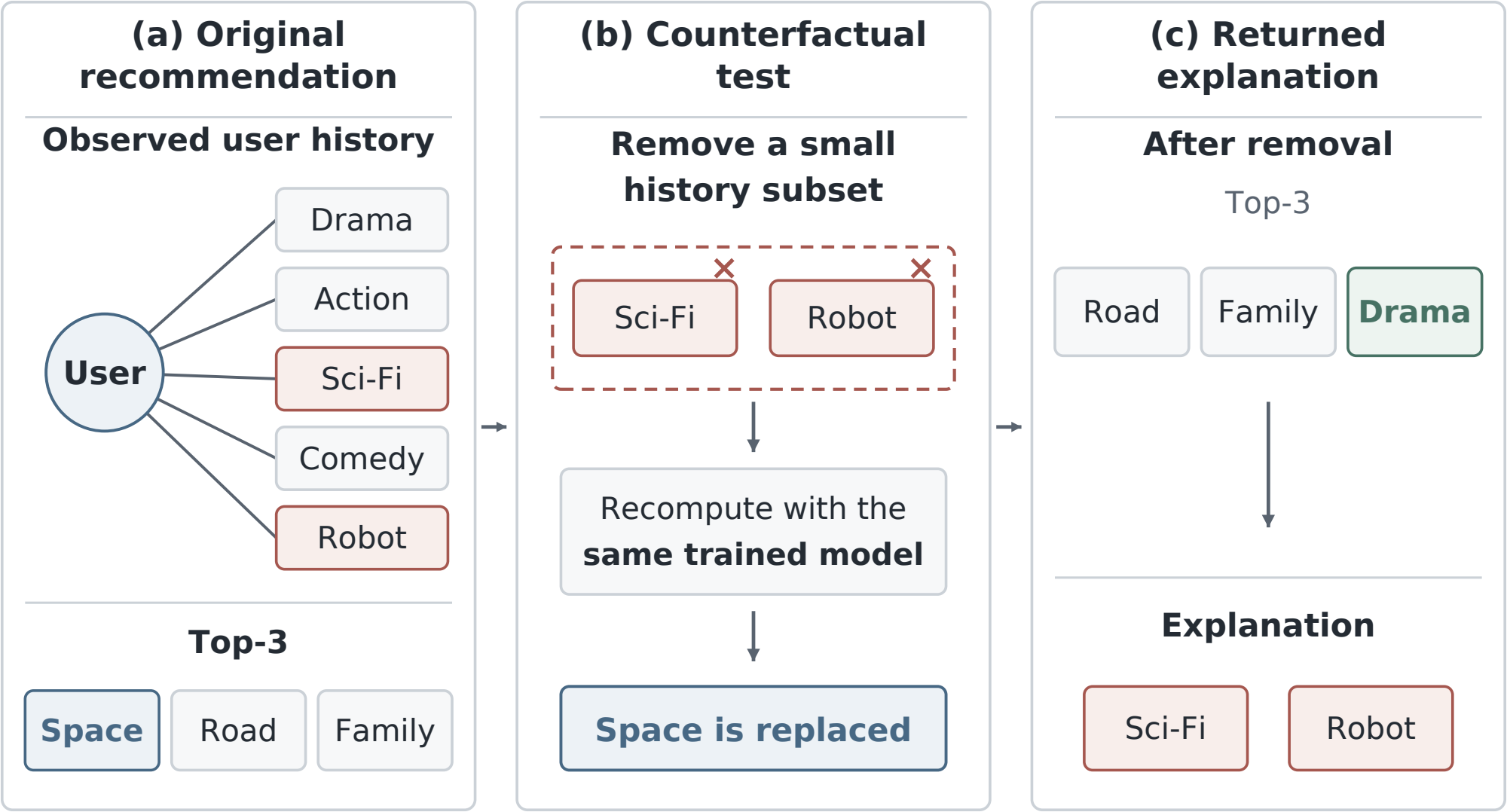

(a) Original recommendation
Observed user history
User
Drama
Action
Sci-Fi
Comedy
Robot
Top-3
Space
Road
Family
(b) Counterfactual test
Remove a small history subset
Sci-Fi
Robot
Recompute with the same trained model
Space is replaced
(c) Returned explanation
After removal
Top-3
Road
Family
Drama
Explanation
Sci-Fi
Robot